EX NO:1	Butterworth IIR Filters
DATE:	**1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.**

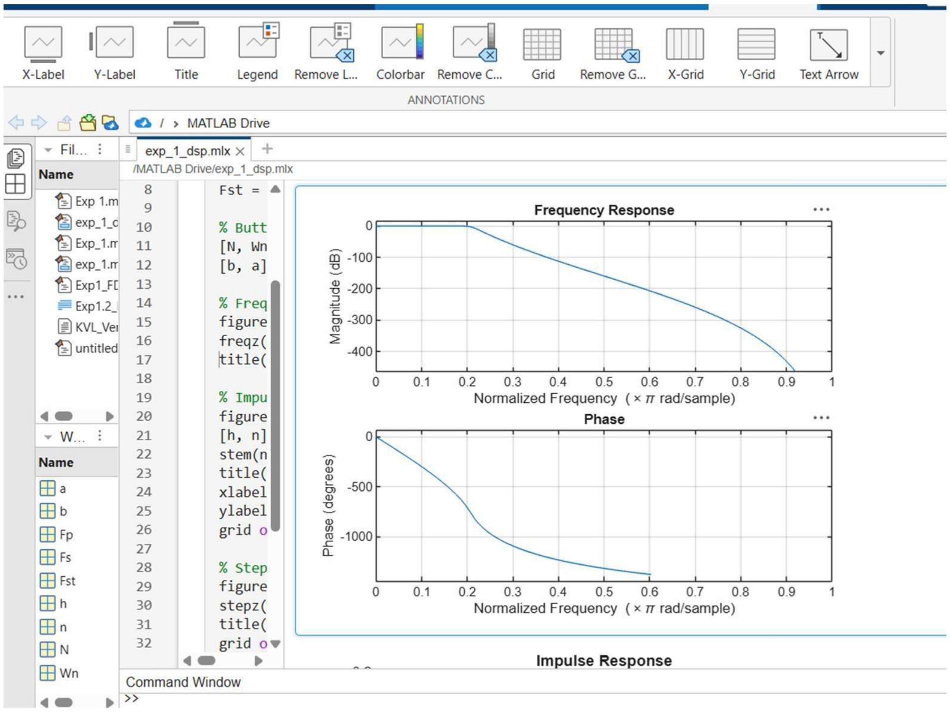
```
clc;
clear;
close all
% Specifications
Fs = 10000;    % Sampling Frequency (Hz)
Fp = 1000;    % Passband Frequency (Hz)
Fst = 1500;    % Stopband Frequency (Hz)
% Butterworth Filter Design
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), 1, 60);
[b, a] = butter(N, Wn);
% Frequency Response
figure;
freqz(b, a);
title('Frequency Response');
% Impulse Response
figure;
[h, n] = impz(b, a, 50);
stem(n, h, 'filled');
title('Impulse Response');
xlabel('Samples');
ylabel('Amplitude');
grid on;
% Step Response
figure;
stepz(b, a);
title('Step Response');
grid on;
% Pole-Zero Plot
figure;
```

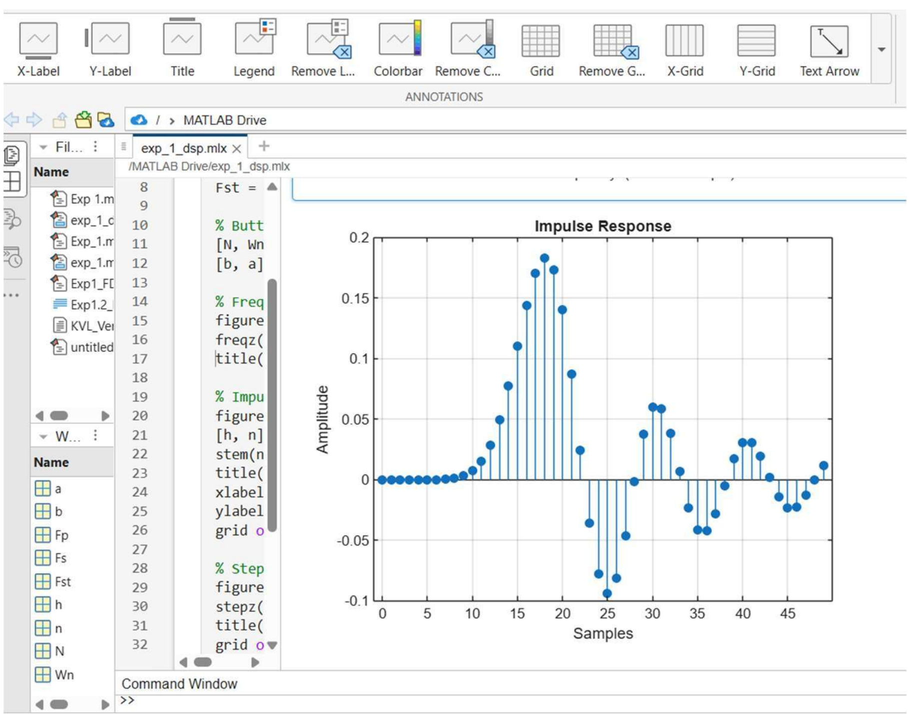
```
zplane(b, a);
title('Pole-Zero Plot');
```

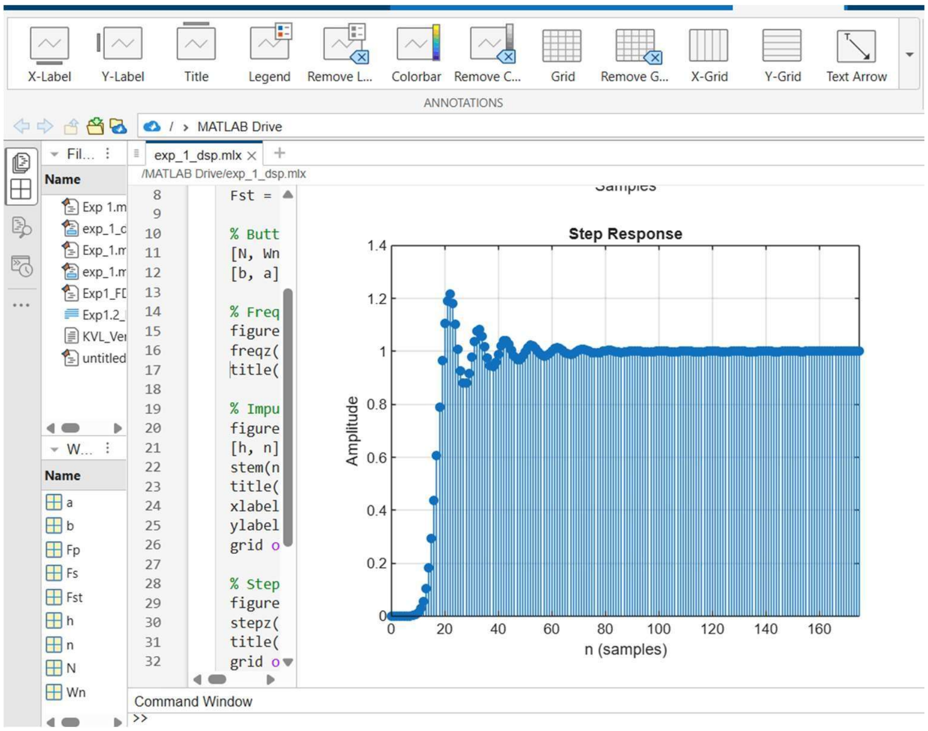
% Display Results

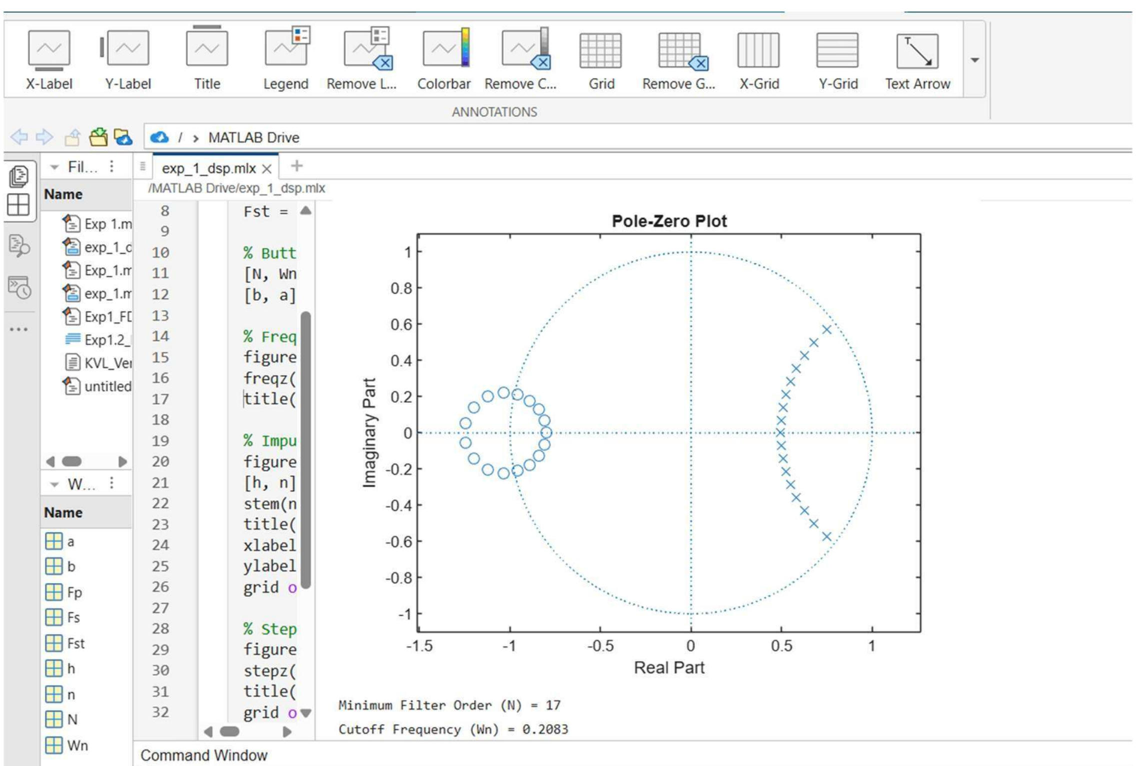
fprintf('Minimum Filter Order (N) = %d\n', N);

printf('Cutoff Frequency (Wn) = %.4f\n', Wn);






X-Label
Y-Label
Title
Legend
Remove L...
Colorbar
Remove C...
Grid
Remove G...
X-Grid
Y-Grid
Text Arrow
ANNOTATIONS
/ > MATLAB Drive
exp_1_dsp.mlx
/MATLAB Drive/exp_1_dsp.mlx
Fst =
% Butt
[N, Wn
[b, a]
% Freq
figure
freqz(
title(
% Impu
figure
[h, n]
stem(n
title(
xlabel
ylabel
grid o
% Step
figure
stepz(
title(
grid o
Name
Exp 1.m
exp_1_c
Exp_1.m
exp_1.m
Exp1_FD
Exp1.2_
KVL_Ver
untitled
a
b
Fp
Fs
Fst
h
n
N
Wn
Step Response
Amplitude
n (samples)
Command Window
>>


X-Label
Y-Label
Title
Legend
Remove L...
Colorbar
Remove C...
Grid
Remove G...
X-Grid
Y-Grid
Text Arrow
ANNOTATIONS
/ > MATLAB Drive
exp_1_dsp.mlx
/MATLAB Drive/exp_1_dsp.mlx
Fst =
% Butt
[N, Wn
[b, a]
% Freq
figure
freqz(
title(
% Impu
figure
[h, n]
stem(n
title(
xlabel
ylabel
grid o
% Step
figure
stepz(
title(
grid o
Name
Exp 1.m
exp_1_c
Exp_1.m
exp_1.m
Exp1_FD
Exp1.2_
KVL_Ver
untitled
a
b
Fp
Fs
Fst
h
n
N
Wn
Pole-Zero Plot
Imaginary Part
Real Part
Minimum Filter Order (N) = 17
Cutoff Frequency (Wn) = 0.2083
Command Window
>>